UML: Class Diagram - Exercise 4 (Flight Booking Agency)

Class diagram

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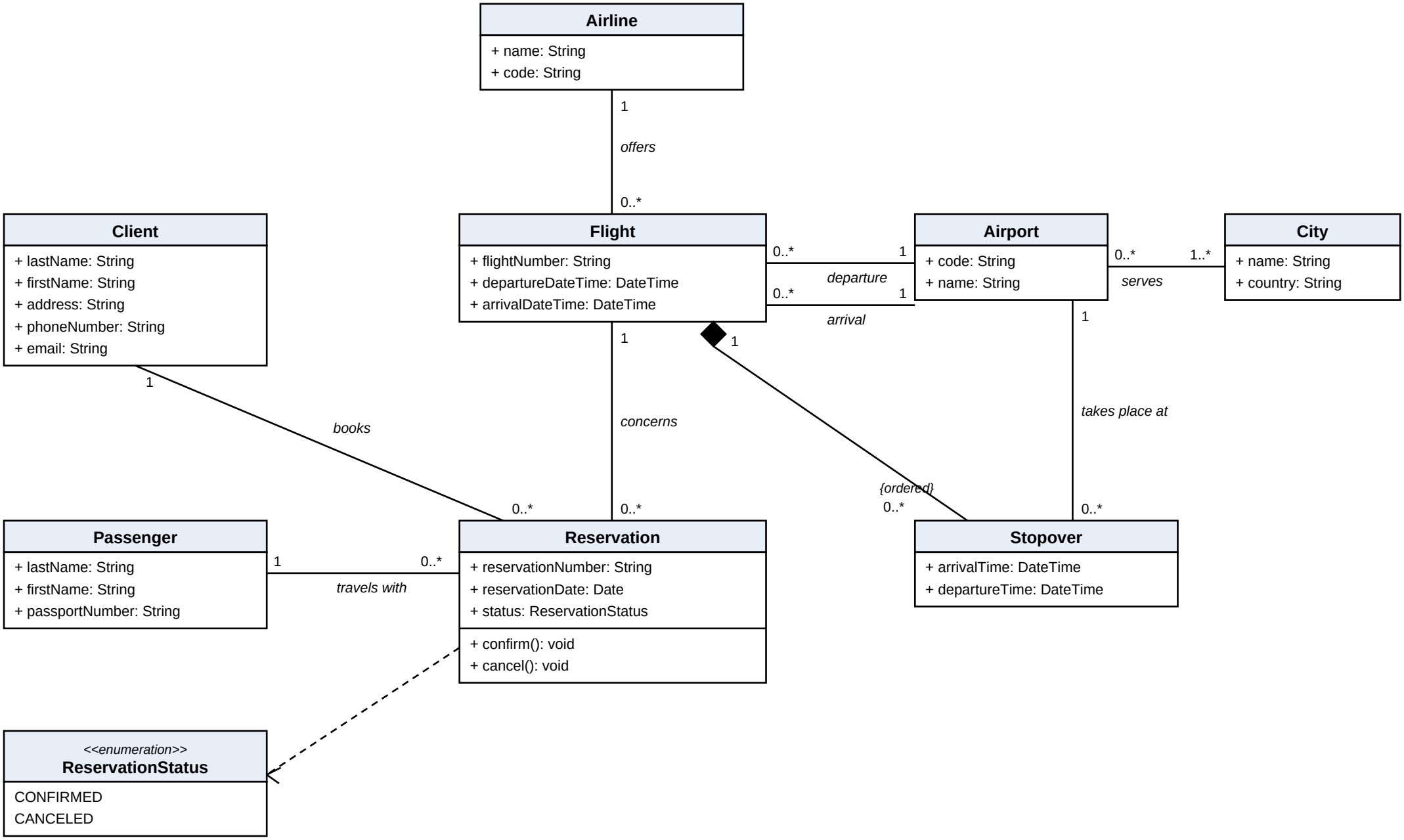
Professor: Imane Fouad - UM6P
Date: October 18, 2026



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Design choices

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Airline and Flight

An airline offers zero or more flights, and each flight is operated by exactly one airline (1 to 0..*).

Client, Passenger and Reservation

The client who books is not necessarily the person who travels, so **Client** and **Passenger** are separate classes. **Reservation** is a full class linked to exactly one Flight and one Passenger, as required by the statement, and a client can make any number of reservations. No direct Client-Flight association is drawn, since it would duplicate the path through Reservation.

Reservation status

The status is typed with the enumeration **ReservationStatus** (CONFIRMED, CANCELED) instead of a String, so only valid values can be stored. The operations confirm() and cancel() change this status.

Departure and arrival airports

A flight has exactly one departure airport and one arrival airport. They are modelled as two distinct associations with the roles *departure* and *arrival*, rather than a single 2..* association, which could not tell them apart. The dates and times are of type DateTime because the statement requires both.

Stopovers

A stopover belongs to a flight: if the flight is deleted, its stopovers disappear. This is a **composition** (filled diamond on the Flight side), with the constraint {ordered} to keep the sequence of stops. Each stopover takes place at one airport and has its own arrival and departure times.

Airport and City

Each airport serves one or more cities (1..*), and a city may be served by several airports (0..*).